

Detailed AWS Training Guide Cloud Foundations & Data Engineering

This guide explains important AWS services used in modern backend systems, data engineering platforms, analytics systems, event-driven architectures, and cloud-native applications.

The goal is not to memorize AWS services individually, but to understand: Why these services exist What problems they solve How they interact in real-world architectures Where companies use them in production

Amazon S3

What is S3?

Amazon S3 (Simple Storage Service) is an object storage service used to store files, logs, images, backups, videos, CSV datasets, machine learning datasets, and application data.

Why was S3 created?

Applications generate huge amounts of files and data. Traditional servers cannot scale infinitely. S3 provides highly durable, scalable cloud storage without managing servers.

Key Concepts: Buckets store files (objects) Objects can be any type of file Virtually unlimited scalability 11 9's durability Supports lifecycle management and versioning

Real-World Example: Netflix stores video content and streaming assets using object storage systems. Analytics companies store terabytes of log files in S3 Data Lakes.

Typical Architecture:

Application Logs → S3 → Athena / Glue Analytics

AWS IAM

What is IAM?

IAM (Identity and Access Management) controls authentication and authorization inside AWS.

Why is IAM important?

Without IAM, anyone could access cloud resources. IAM ensures secure access control.

Core Concepts: Users → individual identities Roles → temporary permissions Policies → permission rules Least privilege principle

Real-World Example: A backend developer may get read-only access to logs but no production DB access. Data engineers may get access only to S3 and Glue.

Important Principle:

Every AWS service ultimately depends on IAM for security.

Amazon VPC

What is VPC?

VPC (Virtual Private Cloud) creates isolated private networks inside AWS.

Why do we need VPC?

Applications require secure networking and traffic isolation. Public internet exposure is dangerous for databases and internal services.

Key Components: Subnets Route Tables Internet Gateway Security Groups NACLs **Real-World Example:**

A company may place: Frontend in public subnet Backend APIs in private subnet Databases in highly restricted subnet **Important Concept:**

VPC provides networking security boundaries.

AWS Glue

What is AWS Glue?

AWS Glue is a serverless ETL (Extract, Transform, Load) platform used for data engineering.

What problem does it solve?

Raw data is usually messy and unusable directly for analytics. Glue transforms raw datasets into cleaned, structured datasets.

Typical ETL Flow:

Raw CSV → Glue ETL → Cleaned Parquet → Analytics

Key Features: Serverless Spark jobs Automatic scaling Integrated with S3 Schema discovery support **Real-World Example:**

E-commerce companies process sales logs daily using ETL pipelines before loading into analytics systems.

AWS Glue Crawlers

What are Glue Crawlers?

Glue Crawlers automatically scan datasets and infer schema information.

Why are Crawlers useful?

Manually defining schemas for large datasets is time-consuming. Crawlers automate metadata discovery.

What do Crawlers detect? Columns Data types Partitions File formats **Real-World Example:**

A crawler scans newly uploaded CSV files in S3 and automatically creates queryable tables.

AWS Glue Data Catalog

What is Glue Data Catalog?

Glue Data Catalog is a centralized metadata repository.

Why do we need metadata?

Analytics systems must know: table names column names data types partition structure **Real-World Example:** Athena queries rely on Glue Catalog metadata to understand datasets stored in S3.

AWS Lambda

What is AWS Lambda?

Lambda is a serverless compute platform that runs code without managing servers.

Key Concept:

Event-driven computing.

Typical Flow:

S3 Upload → Lambda Trigger → Process File

Benefits: No server management Automatic scaling Pay only when code runs **Real-World Example:**

Whenever a user uploads image: Lambda generates thumbnail Stores optimized version in S3

Amazon Kinesis Data Streams

What is Kinesis?

Kinesis Data Streams ingests real-time streaming data continuously.

Why streaming systems?

Some systems generate massive continuous events: Stock prices User clicks IoT sensors Application logs

Real-World Example:

Uber streams real-time location updates from drivers continuously.

Amazon Kinesis Data Firehose

What is Firehose?

Kinesis Firehose automatically delivers streaming data into destinations like: S3 Redshift OpenSearch **Difference**

Between Streams vs Firehose: Streams → custom processing Firehose → managed delivery **Real-World**

Example:

Application logs → Firehose → S3 Data Lake

Amazon Redshift

What is Redshift?

Redshift is a cloud-native analytics data warehouse.

Why data warehouses?

Operational databases are not optimized for heavy analytics queries. Redshift supports large-scale analytical workloads.

Typical Use Cases: Business Intelligence Dashboards Sales analytics Trend analysis **Real-World Example:**

Retail companies analyze years of sales data using Redshift.

Amazon Athena

What is Athena?

Athena is a serverless SQL query engine for querying S3 data directly.

Key Benefit:

No infrastructure setup required.

Typical Flow:

S3 Dataset → Athena SQL Query

Real-World Example:

Data analysts query application logs stored in S3 using SQL.

Amazon MWAA (Managed Airflow)

What is MWAA?

Managed Workflows for Apache Airflow is AWS-managed Airflow orchestration service.

Why orchestration?

Large data pipelines contain multiple dependent steps.

Example Pipeline: Extract data Transform data Load warehouse Trigger reports **Real-World Example:** Daily ETL jobs scheduled automatically at midnight.

AWS Step Functions

What are Step Functions?

Step Functions orchestrate distributed workflows visually.

Why useful?

Modern systems contain multiple microservices and async steps.

Typical Flow:

Upload File → Validate → Process → Notify User

Real-World Example:

Loan approval systems with multiple validation stages.

Amazon EventBridge

What is EventBridge?

EventBridge is an event routing system.

Why event-driven architecture?

Services should communicate asynchronously instead of tightly coupling systems.

Real-World Example:

Order Created Event → Trigger: Email Service Billing Service Shipping Service

AWS DMS

What is DMS?

Database Migration Service migrates databases with minimal downtime.

Why needed?

Companies migrate: on-prem databases legacy systems different DB engines **Real-World Example:**
Migrating MySQL database into AWS Aurora without application downtime.

Complete Real-World AWS Architecture Example

User Activity Logs ↓
Kinesis Data Streams ↓
Kinesis Firehose ↓
Amazon S3 Data Lake ↓
Glue Crawlers discover schema ↓
Glue Data Catalog stores metadata ↓
Athena queries datasets ↓
Redshift stores analytics data

Workflow Orchestration: MWAA + Step Functions + EventBridge

Security Layer: IAM + Secrets Manager + VPC

Key Learning Objective:

Modern cloud systems are combinations of: Storage systems Streaming systems ETL systems Analytics systems
Security systems Workflow orchestration systems Do not memorize AWS services individually.
Understand how they interact to build scalable production systems.